Practical 1: Intelligent Behavior using Simple Rules

1. What is intelligent behavior in AI systems?
   Intelligent behavior refers to the capability of a system to process input, apply logical reasoning, and produce meaningful outputs. It mimics human thinking using predefined rules and helps in automated decision-making.

2. Explain rule-based systems with an example.
   Rule-based systems use IF-THEN rules stored in a knowledge base to make decisions. For example, IF temperature > 30°C THEN turn ON fan. These systems are simple but effective for well-defined problems.

3. What is inference and why is it important?
   Inference is the process of deriving new knowledge from existing facts using logical rules. It is important because it enables systems to make decisions rather than just storing data.

4. Differentiate forward and backward chaining.
   Forward chaining is data-driven and starts from known facts to reach a conclusion, while backward chaining is goal-driven and starts from a goal to verify if it can be achieved.

5. What is a knowledge base?
   A knowledge base is a structured collection of facts and rules used by an AI system. It acts as the core component for reasoning and decision-making.

6. What are advantages of rule-based systems?
   They are simple to design, easy to understand, and useful for solving structured problems. They are widely used in expert systems.

7. What are limitations of rule-based systems?
   They cannot handle uncertainty, lack learning ability, and become complex when the number of rules increases.

8. What is an AI agent?
   An AI agent is an entity that perceives its environment through sensors and acts upon it using actuators to achieve specific goals.

9. What is condition-action rule?
   It is a rule structure where a condition is evaluated, and if true, an action is executed, forming the basis of rule-based systems.

Practical 2: Simple Chatbot

1. What is a chatbot and how does it work?
   A chatbot is a software that interacts with users using text or voice. It works by

processing user input, matching patterns, and generating predefined or learned responses.

2. Explain NLP in chatbot systems.
   Natural Language Processing enables chatbots to understand human language by analyzing syntax and semantics, making interactions more natural.

3. What is pattern matching?
   Pattern matching compares user input with predefined templates to generate responses. It is commonly used in rule-based chatbots.

4. What is intent recognition?
   Intent recognition identifies the purpose of user input, helping the chatbot decide the correct response or action.

5. What are types of chatbots?
   Mainly rule-based chatbots and AI-based chatbots. Rule-based follow fixed logic, while AI chatbots learn from data.

6. What are limitations of simple chatbots?
   They cannot understand context, fail with complex queries, and rely heavily on predefined rules.

7. How can chatbot performance be improved?
   By using machine learning, NLP, and large datasets, chatbots can learn patterns and improve accuracy.

8. What is tokenization?
   It is the process of splitting text into smaller units like words, which helps in text analysis.

9. What are real-world applications?
   Customer support, banking assistants, e-commerce help, and virtual assistants like Siri.

Practical 3: Tautology

1. What is a tautology?
   A tautology is a logical statement that is always true regardless of the truth values of its variables, ensuring logical validity.

2. What is a proposition?
   A proposition is a declarative statement that can either be true or false but not both.

3. Explain truth table.
   A truth table lists all possible combinations of truth values and shows the output of a logical expression for each case.

4. What is contradiction?
   A contradiction is a statement that is always false under all possible conditions.
5. What is logical implication?
   It is a statement of the form IF P THEN Q, where Q depends on P.
6. Why are tautologies important?
   They help verify logical correctness and are used in proofs and digital circuit design.
7. How do you check tautology programmatically?
   By generating all combinations of truth values and evaluating the expression using logical operators.
8. What is Boolean algebra?
   It is a mathematical system dealing with true/false values used in logic and computer science.
9. What are logical operators?
   Operators like AND, OR, NOT, XOR used to combine or modify logical expressions.

Practical 4: Wumpus World

1. Explain Wumpus World environment.
   It is a grid-based AI environment where an agent navigates using logical reasoning to find gold while avoiding dangers.
2. What is an agent?
   The agent is the decision-making entity that uses percepts to safely explore the environment.
3. What are percepts?
   Percepts are sensory inputs like breeze and stench that help the agent detect nearby dangers.
4. What is the goal of the agent?
   The agent aims to find gold and exit safely without falling into pits or encountering the Wumpus.
5. What type of environment is it?
   It is partially observable, deterministic, and sequential.
6. What is logical inference here?
   The agent uses inference to deduce safe and unsafe cells based on percepts.
7. Why is Wumpus World important?
   It helps understand knowledge-based agents and reasoning under uncertainty.

8. What are hazards in Wumpus World?
   Pits and the Wumpus creature are hazards.

Practical 5: Semantic Network

1. What is a semantic network?
   It is a graphical representation of knowledge using nodes (concepts) and edges (relationships).
2. What are nodes and edges?
   Nodes represent objects, while edges define relationships between them.
3. What is inheritance?
   It allows child nodes to inherit properties from parent nodes, reducing redundancy.
4. What is knowledge representation?
   It is the technique of storing structured information for AI processing.
5. What are advantages?
   Easy visualization and intuitive understanding of relationships.
6. What are limitations?
   Complex for large datasets and lacks formal semantics.
7. Where is it used?
   Expert systems, NLP, and AI knowledge bases.
8. What is hierarchy?
   It is a structured arrangement of nodes in levels.

Practical 6: Sample Space using itertools

1. What is sample space?
   It is the set of all possible outcomes of an experiment.
2. What is permutation?
   Arrangement of items where order matters.
3. What is combination?
   Selection where order does not matter.
4. What is itertools?
   Python module for generating combinations and permutations efficiently.

5. How is probability calculated?
   Probability = Favorable outcomes / Total outcomes.

6. What is event?
   A subset of sample space.

7. What is experiment?
   A process producing outcomes.

8. Why use itertools?
   It simplifies combinatorial computations.

Practical 7: Bayesian Network

1. What is a Bayesian Network?
   A Bayesian Network is a probabilistic graphical model that represents variables and their dependencies using nodes and directed edges. It helps in reasoning under uncertainty.

2. What is conditional probability?
   Conditional probability is the probability of an event occurring given that another event has already occurred. It is denoted as P(A|B).

3. What is Bayes' Theorem?
   It is a mathematical formula used to update the probability of a hypothesis based on new evidence. It connects prior and posterior probabilities.

4. What is prior probability?
   Prior probability is the initial belief about an event before considering any new evidence or data.

5. What is posterior probability?
   Posterior probability is the updated probability of an event after taking new evidence into account.

6. What are nodes and edges in Bayesian Network?
   Nodes represent random variables, and edges represent probabilistic dependencies between them.

7. What are applications of Bayesian Networks?
   Used in medical diagnosis, spam filtering, weather prediction, and decision-making systems.

8. What is the advantage of Bayesian Networks?
   They can handle uncertainty and incomplete data effectively.

Practical 8: K-Means Algorithm

1. What is K-Means clustering?
   K-Means is an unsupervised learning algorithm that divides data into K clusters based on similarity. Each cluster is represented by a centroid.

2. How does K-Means algorithm work?
   It initializes centroids, assigns data points to the nearest centroid, and updates centroids iteratively until convergence.

3. What is a centroid?
   A centroid is the average position of all points in a cluster and represents the cluster center.

4. What is the role of K?
   K determines the number of clusters to be formed, and choosing the right K is important for accuracy.

5. What is Euclidean distance?
   It is a distance measure used to calculate how close data points are to centroids.

6. What are advantages of K-Means?
   It is simple, fast, and works well with large datasets.

7. What are limitations of K-Means?
   It is sensitive to initial centroid selection and may give different results for different K values.

8. Where is K-Means used?
   Used in customer segmentation, image compression, and pattern recognition.

Practical 9: KNN (K-Nearest Neighbors)

1. What is KNN algorithm?
   KNN is a supervised learning algorithm used for classification and regression based on nearest neighbors.

2. How does KNN work?
   It calculates the distance between a new data point and existing points, then assigns the class based on majority voting.

3. What is the role of K?
   K defines the number of nearest neighbors considered, affecting model accuracy and decision boundaries.

4. What is distance metric?
   It is used to measure similarity between data points, commonly using Euclidean distance.
5. What are advantages of KNN?
   It is simple, easy to implement, and does not require training phase.
6. What are limitations of KNN?
   It is slow for large datasets and sensitive to noise and irrelevant features.
7. What is the Iris dataset?
   It is a popular dataset used for classification of flower species based on features like petal length and width.
8. Where is KNN used?
   Used in recommendation systems, pattern recognition, and classification tasks.

Practical 10: Neural Network (XOR Problem)

1. What is a neural network?
   A neural network is a computational model inspired by the human brain that processes data using layers of interconnected neurons.
2. What is the XOR problem?
   XOR is a non-linear problem that cannot be solved using a single-layer perceptron, requiring hidden layers.
3. What is an activation function?
   It introduces non-linearity into the network, enabling it to learn complex patterns.
4. What is forward propagation?
   It is the process of passing inputs through the network to compute the output.
5. What is backpropagation?
   It is a learning algorithm used to update weights by minimizing error through gradient descent.
6. What is a loss function?
   It measures the difference between predicted output and actual output.
7. What are weights and bias?
   Weights determine the importance of inputs, while bias helps shift the output.
8. Where are neural networks used?
   Used in image recognition, speech processing, and predictive analytics.